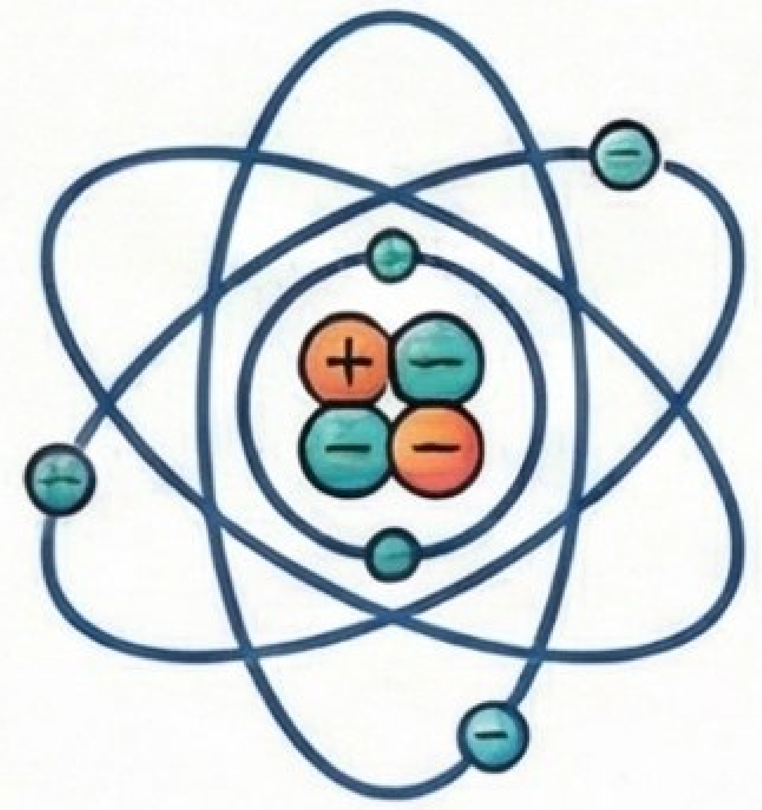
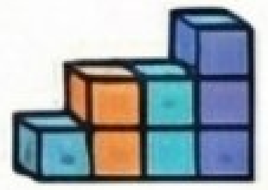
A Visual Guide to Electric Charge & Fields



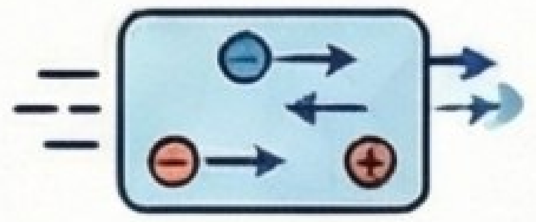
An intrinsic property from electron transfer, exerting electrostatic forces.



Quantization: $Q = ne$

$e = 1.6 \times 10^{-19}$ C

(Total charge = total charge as integer multiple)



Conservation

Charge can neither be created nor destroyed.

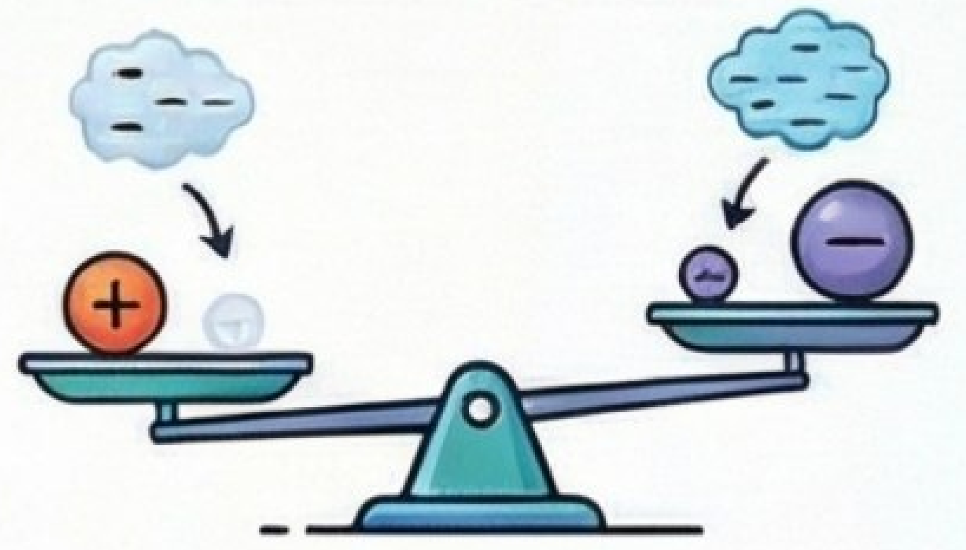
Fundamental Interactions

Like charges repel, opposite charges attract.

The value of an electric charge does not change with the speed of the body.

1. The Basics of Electric Charge

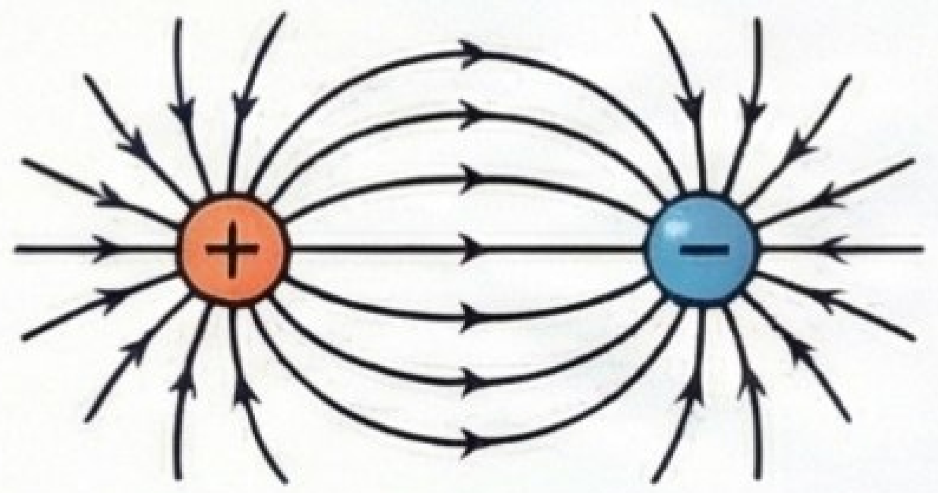
Electron Transfer



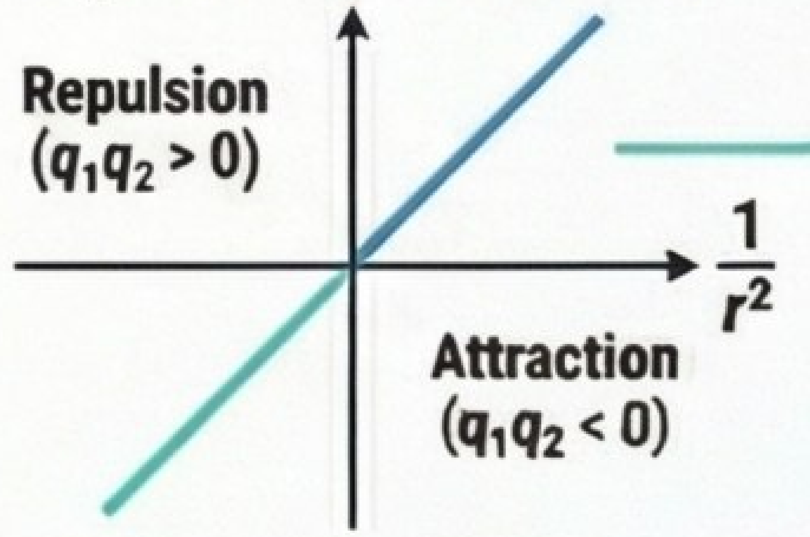
Three Methods of Charging

Friction | Conduction | Induction

Invariance



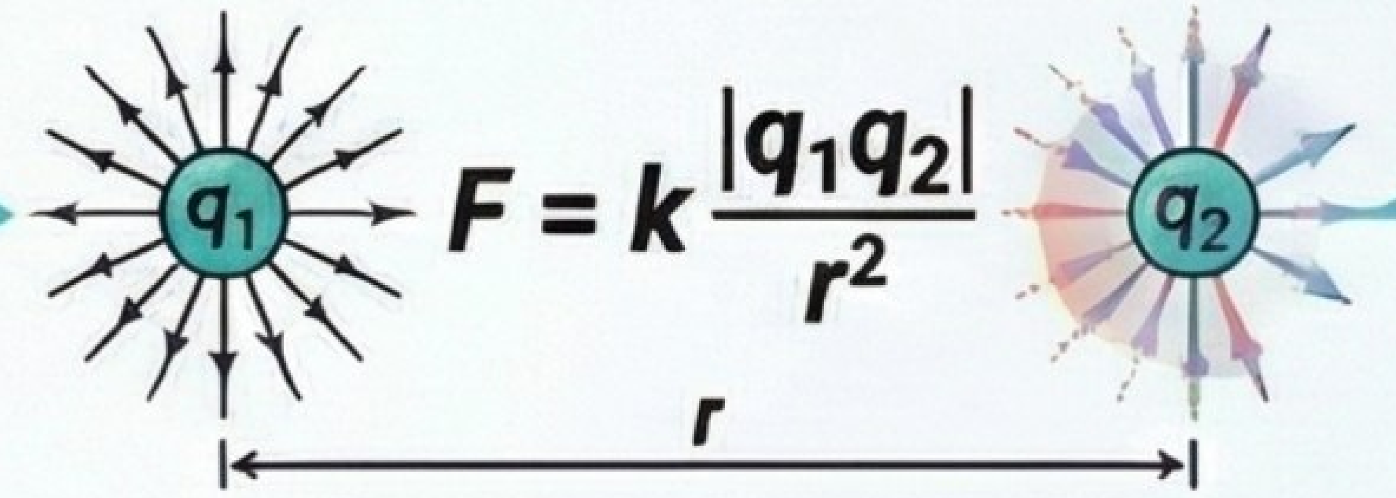
Graph of Force vs. $1/r^2$



Larger $|q_1q_2|$ = Steeper slope

2. Coulomb's Law: The Force Between Charges

Quantifies electrostatic force between stationary point charges.



$$F = k\frac{|q_1q_2|}{r^2}$$

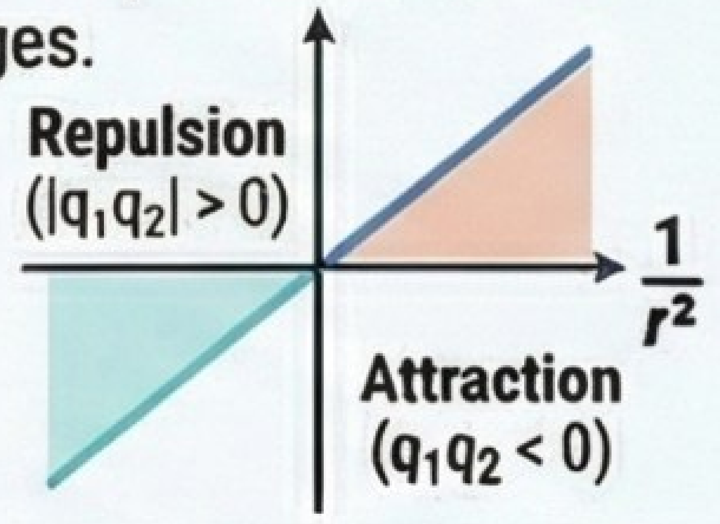
Quantifies electrostatic force between stationary point charges.

$k = 9 \times 10^9$ Nm²/C²

Limitations

- ☑ Point charges
- ☑ Charges at rest
- ☑ Distance > nuclear size

Graph of Force vs. $1/r^2$

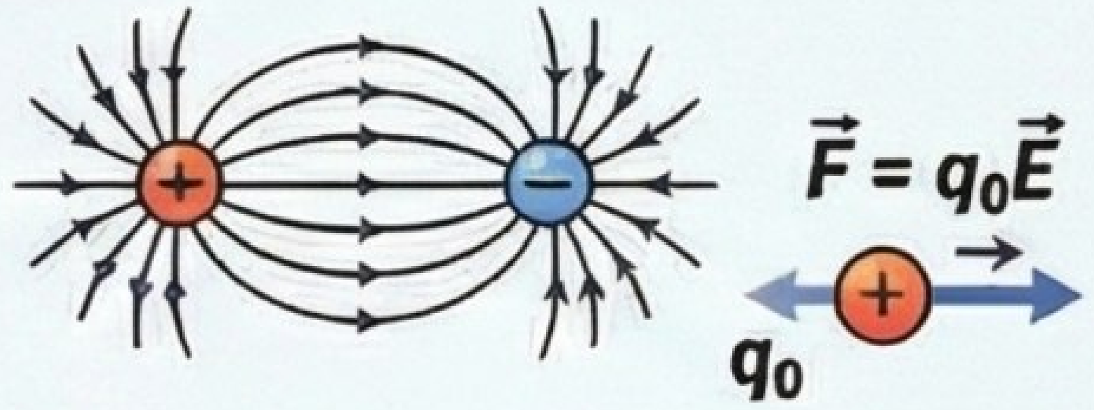


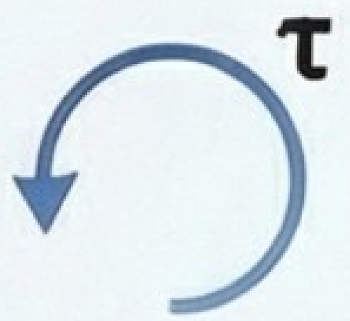
Larger $|q_1q_2|$ = Steeper slope

3. The Electric Field & Field Lines

What is an Electric Field?

The region around a charge where another charge experiences an electrostatic force.





Torque on a Dipole:

$\tau = pE\sin\theta$

Turning force, max at 90°, zero at 0° & 180°

Properties of Electric Field Lines

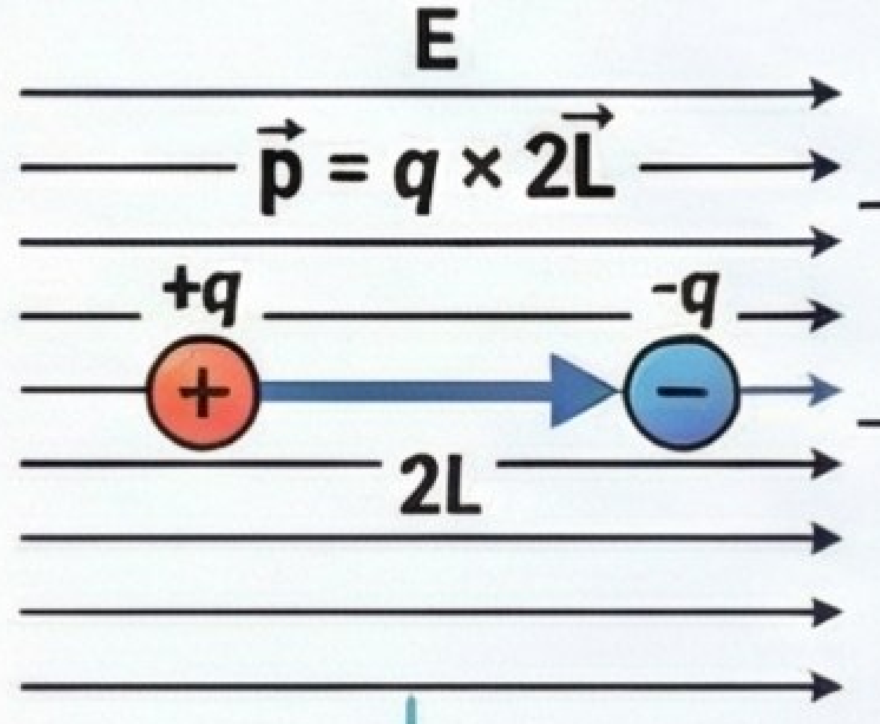
- ☑ Imaginary
- ☑ Start +ve, end -ve
- ☑ Never intersect
- ☑ No closed loops
- ☑ Perpendicular to conductor surface

Potential Energy:

$U = -pE\cos\theta$

Min (-pE, 0°, stable) | Max (+pE, 180°, unstable)

4. Electric Dipoles



Field on Axial Point:

$$E = \frac{1}{4\pi\varepsilon_0}\cdot\frac{2p}{r^3}$$

(stronger, same direction as p)

Field on Equatorial Point:

$$E = \frac{1}{4\pi\varepsilon_0}\cdot\frac{p}{r^3}$$

(weaker, opposite direction to p)

Torque on a Dipole:

$\tau = pE\sin\theta$

Torque is maximum at 90° and zero at 0° & 180°.

Potential Energy:

$U = -pE\cos\theta$

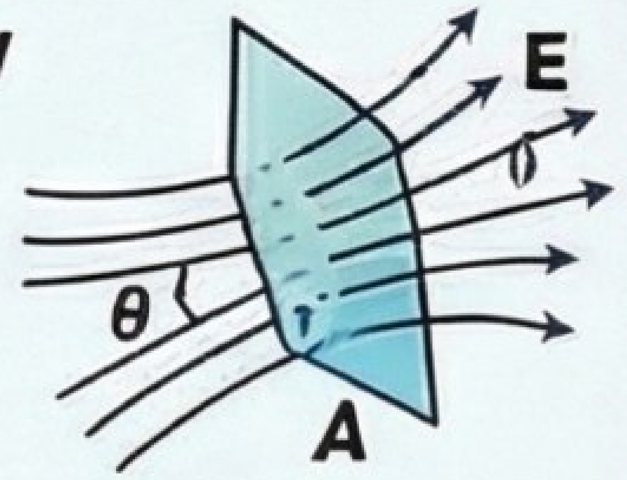
- Min ($-pE$, 0°, stable)
- Max ($+pE$, 180°, unstable)

5. Electric Flux & Gauss's Law

What is Electric Flux (Φ)?

Total electric field lines passing perpendicularly through a surface.

$\Phi = E \cdot A = EA\cos\theta$



Gauss's Law: $\Phi_{closed} = \frac{q_{enclosed}}{\varepsilon_0}$

Charge Distribution	Formula for Electric Field (E)	Key Characteristic
Infinitely Long Wire	$\frac{\lambda}{2\pi\varepsilon_0 r}$	E is inversely proportional to distance (r).
Infinite Plane Sheet	$\frac{\sigma}{2\varepsilon_0}$	E is constant and independent of distance.
Charged Spherical Shell	Outside: $\frac{kQ}{r^2}$ | On Surface: $\frac{kQ}{R^2}$ | Inside: 0	Field is zero inside, maximum on surface, decreases outside.